Sanduni Navodya

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• LinkedIn • GitHub • Portfolio

Software Engineering and Data Science professional with 1.5+ years of hands-on industry experience in Python development, data engineering, AI/ML, healthcare automation, and data-driven applications. Experienced in developing and automating data workflows using Python, Selenium, SQL, Snowflake, and REST APIs, with additional experience in FastAPI, Flask, Docker, Kubernetes, and cloud technologies. Strong background in machine learning, NLP, data pipelines, ETL, database technologies, and AI application development, with proven experience contributing to software and data engineering projects in collaborative environments.

Education

JUNE 2022 – JUNE 2026
BSc. (Hons) in Information Technology specialized in Data Science
Sri Lanka Institute of Information Technology (SLIIT)
Second Class Upper Division | WGPA: 3.54 / 4.00

JANUARY 2012 – DECEMBER 2021
H/Theraputta National School
G.C.E Advanced Level (Physical Science) – 2A's and 1B

Publications

S. Gunathilaka, D. Ranaweera, S. Navodya, D. Himsara, S. Rathnayake, and S. Thelijjagoda, "A Multi-Modal AI System for Epilepsy Diagnosis and Treatment Response," 2026 IEEE International Conference on Electro Information Technology (eIT), 2026, pp. 356–361.
DOI: 10.1109/eIT68936.2026.11670359.

Experience

NOVEMBER 2025 – PRESENT
Associate Software Engineer | Innobot Health | Ethul Kotte, Colombo, Sri Lanka
- Develop and automate end-to-end healthcare data workflows using Python, Selenium, and Snowflake.
- Build and maintain automated data-processing workflows to reduce manual effort and improve operational efficiency.
- Develop reusable Python automation components and wrappers for healthcare workflow processes.
- Perform data validation, troubleshooting, and workflow optimization to improve reliability and consistency.

MAY 2025 – NOVEMBER 2025
Intern Python Developer | Innobot Health | Ethul Kotte, Colombo, Sri Lanka
- Developed healthcare data workflows using Python, Selenium, and Snowflake.
- Supported data pipeline development, workflow orchestration, and process optimization.
- Automated repetitive data processing activities to improve workflow efficiency and reliability.

DECEMBER 2024 – JUNE 2025
Intern Data Engineer | Sri Lanka Telecom PLC | Colombo 01, Sri Lanka
- Contributed to the Racoon AI Project by developing AI and data engineering solutions using Python, LangChain, and JavaScript.
- Developed data-driven application components using Docker, Kubernetes, PostgreSQL, and Redis.
- Deployed software and data engineering solutions using containerized technologies.
- Improved application functionality, scalability, performance, and data management through software and data engineering practices.

Projects

CryGuard – Cry Detection & Environment Monitoring System
- Developed a real-time baby monitoring system using IoT and Machine Learning to detect baby cry conditions.
- Implemented monitoring of temperature, humidity, motion, and light using IoT sensors.
- Integrated live alerts and dashboard updates using ESP32, FastAPI, MongoDB, WebSocket, and React.js.
- Applied Python and Machine Learning techniques to support automated cry-condition detection.

Technologies: Python, Machine Learning, ESP32, IoT Sensors, FastAPI, MongoDB, React.js, WebSocket

Hospital Data Warehouse & Business Intelligence
- Designed a hospital data warehouse using Microsoft SQL Server and a Snowflake Schema.
- Developed ETL workflows using SSIS to extract, transform, cleanse, and load healthcare data.
- Structured fact and dimension tables to support business intelligence reporting.
- Applied T-SQL and data warehousing techniques to organize and process healthcare data.

Technologies: Microsoft SQL Server, SSIS, T-SQL, ETL, Data Warehousing, Snowflake Schema, Business Intelligence

EduPulse – Knowledge Sharing & Learning Platform

- Developed a collaborative learning platform for students to share knowledge and follow structured learning plans.
- Implemented real-time notifications and milestone tracking functionality.
- Integrated secure user authentication using OAuth 2.0.
- Built real-time communication features using WebSocket technology.

Technologies: React.js, Spring Boot, MongoDB, OAuth 2.0, WebSocket

Kings Store – Mobile Phone Recommendation System

- Developed a mobile phone recommendation system using content-based and keyword-based filtering techniques.
- Implemented personalized recommendations based on brand, storage, and price preferences.
- Applied Machine Learning techniques to generate product recommendations.

Technologies: Python, Streamlit, Machine Learning

House Price Prediction System

- Developed a web-based house price prediction application using regression models.
- Implemented property-based prediction using features such as size, structure, and sale details.
- Integrated the machine learning model with a web application using Flask and React.

Technologies: React, Flask, Vite, Machine Learning

Text Summarization and Analysis System

- Implemented NLP models to generate concise summaries of lengthy texts.
- Applied Natural Language Processing techniques to improve information extraction and readability.
- Developed the application using Python and Streamlit.

Technologies: Python, NLP, Streamlit, Machine Learning

GenAI Cold Email Generator

- Developed a GenAI application that generates personalized cold emails from job listings.
- Implemented job listing extraction from company career pages.
- Integrated portfolio information stored in a vector database to generate personalized content.
- Applied Groq and LangChain technologies to implement the AI workflow.

Technologies: Python, Groq, LangChain, Streamlit, Vector Database

Skills

Programming: Python, SQL, Java, JavaScript, R
Data Science & AI: Machine Learning, Artificial Intelligence, Natural Language Processing, Data Science, Data Analysis, Generative AI
Data Engineering: Data Pipelines, ETL, Data Warehousing, Snowflake, SSIS, T-SQL
Backend & Automation: FastAPI, Flask, REST APIs, API Integration, Selenium
Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Redis
Cloud & DevOps: Microsoft Azure, AWS, Docker, Kubernetes, Linux
AI & Development Tools: LangChain, Streamlit, Jupyter Notebook, Git
Frontend & Web: React.js, Tailwind CSS, MERN Stack

Awards & Certifications

- Dean's List for Year 4 Semester 2 for maintaining a GPA above 3.7.
- Merit Award Winner – CODEFEST 2024 (Algothon – Tertiary Category), organized by SLIIT.
- Most Popular Team Award – Mini Hackathon 2024, organized by the Microsoft Club of SLIIT in collaboration with FCSC.
- AWS Academy Graduate – AWS Academy Cloud Foundations.
- AI/ML Engineer – Stage 1 Certification.

References

Prof. Samantha Thelijjagoda
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Lecturer
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